Time Harmonic Fields, Complex Permittivity, and Helmholtz Equations

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Lecture Objective

- To introduce the concepts of complex permittivity, admittivity and impedivity as the relevant properties of a medium for a time-harmonic electromagnetic wave.
- To derive the electromagnetic Helmholtz equations from the time-harmonic Maxwell's equations.

Linearity of Maxwell's Equations

- Maxwell's equations are linear differential equations provided the media involved are all linear.
- The waves we deal with in EEE 341 are assumed to exist in simple media.

Consequences of Linearity …

- The property of linearity allows us to make use of the concept of phasors.
- Any arbitrary function of time can be written as a superposition of sinusoids – using Fourier series (for periodic functions) or Fourier transforms (for non-periodic functions).

Time-Harmonic Fields

- If the source of the electromagnetic field is a sinusoidal function of frequency, then the resulting electromagnetic field will be sinusoids of the same frequency, assuming all media are linear.
- “Time-harmonic” is a synonym for “sinusoidal”.

Phasor Representation of Time-Harmonic Vector Field

$$\vec{E}(\vec{R},t) = \mathrm{Re}\left\{\tilde{\vec{E}}(\vec{R})e^{j\omega t}\right\}$$

Phasor (frequency domain) representation of electric field

Sinusoidal function of time

Phasor Representation of Time-Harmonic Vector Field

$$\tilde{\vec{E}}(\vec{R}) = \hat{a}_E(\vec{R})E(\vec{R})\exp\left(j\theta_E(\vec{R})\right)$$

Amplitude

Phase

$\hat{a}_E(\bar{R})$ is a unit vector that gives the direction of the electric field at any location

Time-Harmonic Maxwell's Equations (1 of 2)

- Replace each field quantity with its phasor representation, e.g.,

$$\vec{E}(\vec{R},t) \leftarrow \tilde{\vec{E}}(\vec{R})$$

- Replace time derivatives with multiplication by $j\omega$, e.g.,

$$\frac{\partial}{\partial t}\vec{E}(\vec{R},t) \leftarrow j\omega\tilde{\vec{E}}(\vec{R})$$

Time-Harmonic Maxwell's Equations (2 of 2)

$$\nabla \times \tilde{\vec{E}} = -j\omega \tilde{\vec{B}} \qquad \nabla \cdot \tilde{\vec{D}} = \rho$$

$$\nabla \times \tilde{\vec{H}} = \tilde{\vec{J}}_i + \tilde{\vec{J}}_c + j\omega \tilde{\vec{D}} \qquad \nabla \cdot \tilde{\vec{B}} = 0$$

Here, we have separated the current density into two terms: the impressed current (source of electromagnetic field) and the conduction current (reaction to the electromagnetic field).

Time-Harmonic Maxwell's Equations in a Source-Free Medium

- Electromagnetic fields arise from the sources but exist beyond the region containing the sources.
- The field existing in a source-free region must obey source-free Maxwell's equations:

$$\nabla \times \tilde{\vec{E}} = -j\omega \tilde{\vec{B}} \qquad \nabla \cdot \tilde{\vec{D}} = 0$$
$$\nabla \times \tilde{\vec{H}} = \tilde{\vec{J}}_c + j\omega \tilde{\vec{D}} \qquad \nabla \cdot \tilde{\vec{B}} = 0$$

Time-Harmonic Maxwell's Equations in a Source-Free Simple Medium

- For a simple medium, we have

$$\vec{D} = \varepsilon\vec{E} \quad \vec{B} = \mu\vec{H} \quad \vec{J}_c = \sigma\vec{E}$$

- Maxwell's equations become

$$\boxed{\begin{aligned} &\nabla \times \tilde{\vec{E}} = -j\omega\mu\tilde{\vec{H}} \qquad && \nabla \cdot \tilde{\vec{E}} = 0 \\ &\nabla \times \tilde{\vec{H}} = (\sigma + j\omega\varepsilon)\tilde{\vec{E}} \qquad && \nabla \cdot \tilde{\vec{H}} = 0 \end{aligned}}$$

Complex Permittivity (1 of 2)

- We can rewrite Ampere's law as

$$\nabla \times \vec{\tilde{H}} = (\sigma + j\omega\varepsilon)\vec{\tilde{E}} = j\omega\left(\varepsilon - j\frac{\sigma}{\omega}\right)\vec{\tilde{E}}$$

- And define complex permittivity as

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega}$$

Complex Permittivity (2 of 2)

- In undergraduate textbooks, it is common to assume that ε is purely real.
- Unfortunately, this treatment is inadequate and leads to misleading conclusions for lossy dielectric materials.
- We will deviate a bit from our textbook here and use a more sophisticated treatment of complex permittivity.

Electromagnetic Properties of Materials (1 of 2)

- In the static ($\omega = 0$) case, a material is completely characterized for Maxwell's equations by three real numbers: ε, μ, and σ.
- The permittivity (ε) of the material is due to electric dipoles within the material (valence band electrons).
- The permeability (μ) of the material is due to magnetic dipoles within the material (mainly, electron spin).
- The conductivity (σ) is due to free electrons in the material (conduction band electrons).

Electromagnetic Properties of Materials (2 of 2)

- For the time-harmonic ($\omega > 0$) case, the response of electric and magnetic dipoles in a material can experience phase lag relative to the excitation.
- This phase lag represents a loss mechanism, and means that the permittivity and permeability of the material need to be represented as complex numbers:

$$\varepsilon = \varepsilon' - j\varepsilon''$$
$$\mu = \mu' - j\mu''$$

Impedivity and Admittivity of the Medium

- **Impedivity** and **admittivity** of the medium have dimensions of impedance per unit length and admittance per unit length.

$$z(\omega) = j\omega\mu = \omega\mu'' + j\omega\mu'$$

$$y(\omega) = \sigma + j\omega\varepsilon = \sigma + \omega\varepsilon'' + j\omega\varepsilon'$$

Time-Harmonic Maxwell's Equations in a Source-Free Simple Medium

- In terms of impedivity and admittivity:

$$\nabla \times \tilde{\vec{E}} = -z(\omega)\tilde{\vec{H}} \qquad (1)$$

$$\nabla \times \tilde{\vec{H}} = y(\omega)\tilde{\vec{E}} \qquad (2)$$

$$\nabla \cdot \tilde{\vec{E}} = 0 \qquad (3)$$

$$\nabla \cdot \tilde{\vec{H}} = 0 \qquad (4)$$

Question

- What are the values (including units) for the impedivity and admittivity of free space at a frequency of 915 MHz?

Electromagnetic Helmholtz Equation from Maxwell's Equations (1 of 2)

$\nabla \times (1)$:

Using (2)

$$\nabla \times \nabla \times \tilde{\vec{E}} = -z\left(\nabla \times \tilde{\vec{H}}\right) = -yz\tilde{\vec{E}}$$

Vector identity:

Using (3)

$$\nabla \times \nabla \times \tilde{\vec{E}} = \nabla\left(\nabla \bullet \tilde{\vec{E}}\right) - \nabla^2 \tilde{\vec{E}} = -\nabla^2 \tilde{\vec{E}}$$

Helmholtz Equation for Electric Field

$$\boxed{\nabla^2 \tilde{\vec{E}} - \gamma^2 \tilde{\vec{E}} = 0}$$

$$\gamma = \sqrt{yz} = \alpha + j\beta$$

Electromagnetic Helmholtz Equation from Maxwell's Equations (2 of 2)

$\nabla \times (2)$:

Using (2)

$$\nabla \times \nabla \times \tilde{\vec{H}} = y\left(\nabla \times \tilde{\vec{E}}\right) = -yz\tilde{\vec{H}}$$

Vector identity:

Using (3)

$$\nabla \times \nabla \times \tilde{\vec{H}} = \nabla\left(\nabla \bullet \tilde{\vec{H}}\right) - \nabla^2 \tilde{\vec{H}} = -\nabla^2 \tilde{\vec{H}}$$

Helmholtz Equation for Electric Field

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Propagation, Attenuation and Phase Constants

$$\gamma = \alpha + j\beta$$

Propagation constant (m^{-1})

Attenuation constant (nepers/m)

Phase constant (radians/m)

$$\alpha = \mathrm{Re}(\gamma)$$

$$\beta = \mathrm{Im}(\gamma)$$

Summary

- The concepts of complex permittivity, impedivity and admittivity provide a simple way to represent the electromagnetic properties of a medium at a given frequency.
- Electromagnetic wave behavior arising from a time-harmonic source is described by solutions to the electromagnetic Helmholtz equation.

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